

MP flows project

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Problem with $\Omega = 0$

With $\Omega = 0$, we get

$$\frac{dh}{dt} = -\frac{1}{3}h^3 \sin(\theta) \quad \frac{d\theta}{dt} = -h^2 \cos(\theta)$$

By the chain rule we have that

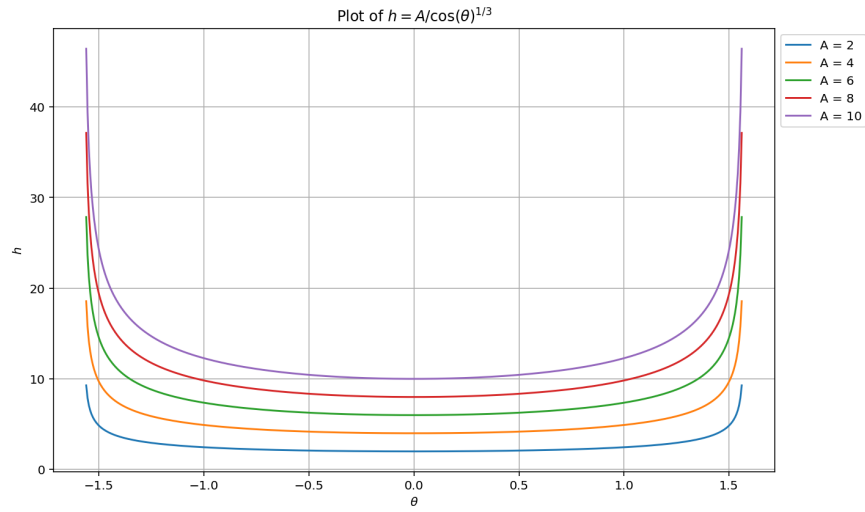
$$\frac{dh}{d\theta} = \frac{1}{3}h \left(\frac{\sin(\theta)}{\cos(\theta)} \right)$$

Solving this by separation of variables we get

$$h^3 \cos(\theta) = A$$

Where A is an arbitrary real constant. Since h cannot be negative (fluid cannot go into the cylinder), and $\theta \in [-\pi/2, \pi/2]$, we have that $A > 0$. Thus, we have

$$h = \frac{A}{\cos(\theta)^{1/3}}$$



Note then that A is dependent on θ_0 and h_0 , so the θ is implicitly considered in calculation by A . With h_0 independent of θ_0 , we have that for each θ_0 , its corresponding A value is given by $A = h_0 \cos(\theta_0)$. There is need to be careful here in interpretation, each h_0 takes a particular constant, given by its starting angle, since each value of h_0 is tracking a particular particle of the fluid surface with evolution given by characteristics. Since A is an arbitrary real constant, and positive, we are free to do substitutions of A to different powers whenever is convenient in calculations for simplification purposes. Plugging back into characteristics, we have

$$\begin{aligned}\frac{dh}{dt} &= -\frac{1}{3}h^3(1 - \cos^2(\theta))^{1/2} \\ &= -\frac{1}{3}h^3(1 - (A^2/h^6))^{1/2} \\ &= -\frac{1}{3}(h^6 - A^2)^{1/2}\end{aligned}$$

So then

$$h(t) = \int \frac{dx}{(h^6 - A^2)^{1/2}}$$

Note that by the generalised binomial theorem, since $h > A$

$$\frac{dh}{dt} = -\frac{1}{3} \sum_{k=0}^{\infty} (-1)^k \binom{1/2}{k} h^{6(1/2-k)} A^{2k}$$

Where

$$\binom{1/2}{k} = \frac{(1/2)^k}{k!} \quad (1/2)^k = \prod_{n=0}^{k-1} (1/2 - n)$$

Note that

$$\binom{1/2}{k} = \frac{(1/2)(1/2-1)(1/2-2) \cdots (1/2-(k-1))}{k!}$$

So then, for k even, there are an even number of negative terms in the numerator, so the expression $(-1)^k \binom{1/2}{k}$ is positive for even k . For odd k , $(-1)^k = -1$ and there are an odd number of terms in the numerator so $\binom{1/2}{k} < 0$. Thus the expression $(-1)^k \binom{1/2}{k}$ is positive for all k . So then

$$\frac{dh}{dt} = -\frac{1}{3} \sum_{k=0}^{\infty} \left| \binom{1/2}{k} \right| h^{3-6k} A^{2k}$$

Looking at the ratio between consecutive terms

$$a_{k+1} = a_k \cdot \frac{A^2(2k-1)}{h^6(2k+2)}$$

Note that $2k + 2 > 2k - 1$ and $h^3 > A$, $A > 0 \implies h^6 > A^2$, so

$$\frac{2k-1}{2k+2} < 1 \quad \frac{A^2}{h^6} < 1$$

So

$$\frac{a_{k+1}}{a_k} < 1$$

For all k . With $a_0 = h^3$ we get that

$$a_k = h^3 \left(\frac{A^2(2k-1)}{h^6(2k+2)} \right)^k$$

The sum can thus be rewritten, so

$$\frac{dh}{dt} = -\frac{1}{3}h^3 \sum_{k=0}^{\infty} \left(\frac{A^2(2k-1)}{h^6(2k+2)} \right)^k$$

As $k \rightarrow \infty$ the terms of the sum approach

$$a_k \approx \left(-\frac{1}{6}h^3 \right) \frac{A^{2k}}{h^{6k}} = -\frac{A^{2k}}{6h^{6k-3}} \approx \frac{A^{2k}}{6h^{6k}}$$

As $2k + 2 \approx 2k - 1$ and $6k - 3 \approx 6k$ for large k . This approaches zero, so eventually for some k we should have that higher order terms we can take to be negligible, but not sure how I should choose such a k .

Desmos seems to show that $x \approx 1.3$ is maximum point so that

$$f(x) = \left(\frac{2x-1}{2x+2} \right)^x$$

is decreasing, and that $f(1) = f(2)$. The derivative is very messy so I will trust for now that no more stationary points appear. Since $(A^2/h^6)^k$ is also decreasing with k , we have that the maximum value in the series besides $k = 0$ is at $k = 1$. The partial sums are thus bounded by above

$$s_n = \sum_{k=0}^{\infty} \frac{A^2(2k-1)}{h^6(2k+2)}, \quad 1 < s_n < 1 + n \left(\frac{A^2}{4h^6} \right)$$

The error is

$$1 + n \left(\frac{A^2}{4h^6} \right) - s_n = (n-1) \frac{A^2}{4h^6} + \mathcal{O}\left(\frac{A^4}{h^{12}}\right)$$

If we ignore $\mathcal{O}(A^4/h^{12})$, then we get that

$$s_n = 1 + \frac{A^2}{4h^6}$$

Note that from before we have that $A^2/h^6 = \cos^2(\theta)$. This averages out to

$$\mathbb{E}[\cos^2(\theta)] = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \theta \cos^2(\theta) d\theta = 2 \int_0^{\frac{\pi}{2}} \theta \cos^2(\theta) d\theta = \frac{\pi^2 - 4}{8} \approx 0.183$$

Thus each the expectation of the ratio between each term is approximately 0.183. This forms a geometric series for an approximate expectation of error in terms of the second term

$$\mathbb{E}[\mathcal{O}(\frac{A^4}{h^{12}}) \cdot \frac{4h^6}{A^2}] = \sum_{n=0}^{\infty} (\frac{\pi^2 - 4}{8})^{n+1} = \frac{\pi^2 - 4}{12 - \pi^2} \approx 2.755$$

Thus maybe for some $1 < \alpha \leq 2.755$, a better approximation for s_n is

$$s_n \approx 1 + (1 + \alpha) \frac{A^2}{4h^6}$$

So, ignoring constants, approximating dh/dt with the first two terms, we get

$$\frac{dh}{dt} = h^3 + \frac{1}{h^3} \implies \int \frac{h^3}{h^6 + 1} dh = t$$

We thus want to evaluate the integral (but note, negative ignored, so by what we discussed last time, perhaps this is only valid for $-\pi/2 < \theta \leq 0$?)

$$I = \int \frac{x^3}{x^6 + 1} dx$$

I found this website integral-calculator.com that very nicely solved the integral and showed all steps. It used a nice substitution and factorization to solve it without (explicitly) invoking complex roots. The steps seem correct, and I don't think there are any domain issues. Thus

$$t = \frac{1}{12} \left[\ln(h^4 - h^2 + 1) + 2 \left(\sqrt{3} \arctan\left(\frac{2h^2 - 1}{\sqrt{3}}\right) - \ln(h^2 + 1) \right) \right] + C$$

With $t = 0$, $h = h_0$ (since we have a uniform starting profile for h . If we didn't, then this would be a function of θ),

$$C = 2\sqrt{3} \arctan\left(\frac{2h_0^2 - 1}{\sqrt{3}}\right) + \ln\left(\frac{(h_0^2 + 1)^2}{h_0^4 - h_0^2 + 1}\right)$$

If we let $h_0 = 1$ then $C = \ln(2)$ (which we should be allowed to do no problem).

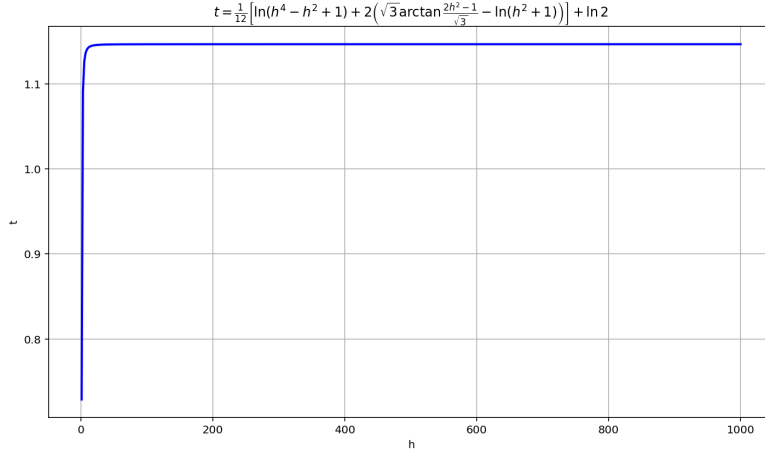
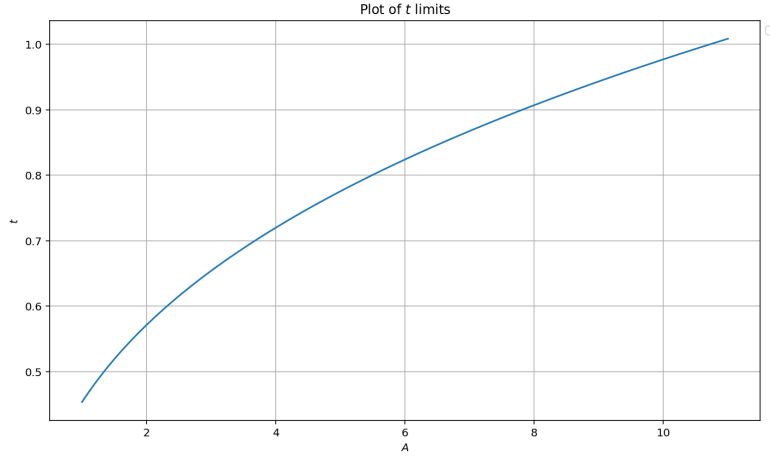


Figure 1: Sampled for h from 1 to 1000

As we can see, it appears that there is an asymptote in t , so system completely drains of fluid in finite time? As $h \rightarrow \infty$, $t \rightarrow \sqrt{3}\pi/12 + C$ as the $\ln$ terms go to 0 and the arctan term goes to $\pi/2$, and C is determined by h_0 . We left out A in our integration steps, but different values of A will lead to different limits for t . With A considered, redoing the diff eq we get $t \rightarrow \sqrt{3}\pi/12A^{1/3}$



Note that

$$I = \int \frac{1}{(h^6 - A^6)^{1/2}} dh = \frac{1}{i} \int \frac{1}{(A^6 - h^6)^{1/2}} dh$$

So then, taking the modulus of both, we have

$$|I| = \left| \frac{1}{i} \int \frac{1}{(A^6 - h^6)^{1/2}} dh \right|$$

Supposing that the result of I is entirely real (might not be) and positive ($t, h > 0$), then $I = |I|$ and thus we can analyse this integral instead

$$\begin{aligned} \int_0^A \frac{1}{(A^6 - h^6)^{1/2}} dh &=_{h=AH} \frac{1}{A^2} \int_0^1 \frac{1}{(1 - H^6)^{1/2}} dH \\ &= \frac{1}{A^2} [H_2F_1(1/6, 1/2, 7/6, H^6)]_0^1 \\ &= \frac{1}{A^2} \frac{\Gamma(\frac{7}{6})\Gamma(\frac{1}{2})}{\Gamma(1)\Gamma(\frac{3}{2})} \\ &= \frac{1}{3} \Gamma\left(\frac{1}{6}\right) \end{aligned}$$

The lower bound $h = 0$ not sure that makes sense since h doesn't start at 0. I suppose what we want to look at is

$$I = \int_{h_0}^A \frac{1}{(A^6 - h^6)^{1/2}} dh = \frac{1}{A^2} \left(\frac{1}{3} \Gamma\left(\frac{1}{6}\right) - h_0 \cdot {}_2F_1(1/6, 1/2, 7/6, H_0^6) \right)$$

Where $h_0 = AH_0$, since $I = t(A) - t(h_0)$, $t(h_0) = 0$ so $I = t(A) = t_A^*$. Note we have that

$${}_2F_1(1/6, 1/2, 7/6, H^6) = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^{\bar{k}} \zeta_k(H^6)$$

Where

$$\begin{aligned} \sum_{k=0}^{\infty} \zeta_k(z) &= {}_2F_1(1/6, 1, 7/6, z) \\ &= \frac{1}{12z^{\frac{1}{6}}} \left(2\sqrt{3} \arctan\left(\frac{1 - z^{\frac{1}{3}}}{\sqrt{3}z^{\frac{1}{6}}}\right) + \log\left(\frac{z^{\frac{1}{3}} + z^{\frac{1}{6}} + 1}{z^{\frac{1}{3}} - z^{\frac{1}{6}} + 1}\right) + 2 \log\left(\frac{1 + z^{\frac{1}{6}}}{1 - z^{\frac{1}{6}}}\right) \right) \end{aligned}$$

If $z^{1/6} > 0$.

$$\left(\frac{1}{2}\right)^{\bar{k}} = 1/2(1/2 + 1) \cdots (1/2 + (k - 1)), \quad k > 0$$

For some k , note that

$$\left(\frac{1}{2}\right)^{\bar{k}} > \frac{1}{2}(n - 1)!$$

So the hypergeometric is bounded from below by

$$\zeta_0(H^6) + \frac{1}{2} \sum_{k=1}^{\infty} (n - 1)! \cdot \zeta_k(H^6)$$

Note then that the effect of $(1/2)^{\bar{k}}$ will result in something larger than the sum of ζ_k , so we can bound t_A^* from above by

$$\frac{1}{A^2} \left(\frac{1}{3} \Gamma \left(\frac{1}{6} \right) - \frac{h_0}{12H_0} \left(2\sqrt{3} \arctan \left(\frac{1-H_0^2}{\sqrt{3}H_0} \right) + \log \left(\frac{H_0^2 + H_0 + 1}{H_0^2 - H_0 + 1} \right) + 2 \log \left(\frac{1+H_0}{1-H_0} \right) \right) \right)$$

Note that since $H_0 > 1$, we have complex numbers in this. But since we are looking at the modulus of our integral times $1/i$, we won't end up with complex numbers. Then, we have the journey from $h = A$ to $h = \infty$,

$$\int_A^\infty \frac{1}{(h^6 - A^6)} dh = \frac{1}{A^2} \int_1^\infty \frac{1}{(H^6 - 1)^{1/2}} dH = \frac{1}{6A^2} \frac{\sqrt{\pi} \Gamma(\frac{1}{3})}{\Gamma(\frac{5}{6})}$$

So, adding the two integrals together will give the total blowup time for h starting in $\pi/2 \leq h \leq 0$? But we integrated the differential equation non-negative, so isn't

Note that from the second characteristic we find

$$A^2 \int \cos(\theta)^{-1/3} d\theta = t$$

Analysing this integral, note that

$$\begin{aligned} I &= \int_0^{\pi/2} \cos(\theta)^{-1/3} d\theta \\ [u=\sin(\theta)] &= \int_0^1 (1-u^2)^{-2/3} du \\ [t=u^2] &= 2 \int_0^1 t^{1/2} (1-t)^{-2/3} dt \\ &= B\left(\frac{3}{2}, \frac{5}{3}\right) \end{aligned}$$

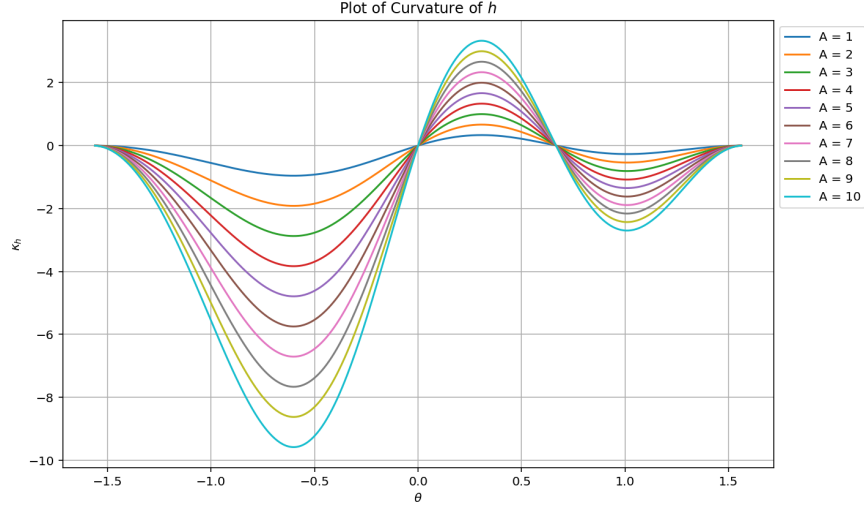
Not exactly sure how to interpret this or what it could mean.

Curvature

I thought looking at curvature might be nice, with $h = \sec(\theta)$,

$$R_h = \frac{(\sec^2(\theta) + (\tan(\theta) \sec(\theta))^2)^{\frac{3}{2}}}{\sec(\theta)[\tan(\theta)(2 - \tan(\theta) \sec(\theta)) + \sec(\theta)(1 - \sec^2(\theta))]}$$

So $\kappa_h = R_h^{-1}$.



κ_h is always negative except for a small region between $\theta = 0$ and the next root of κ_h , which desmos gives as 0.66624 (not sure if this is a significant number). We also have roots at $-\pi/2, \pi/2$. In terms of h ,

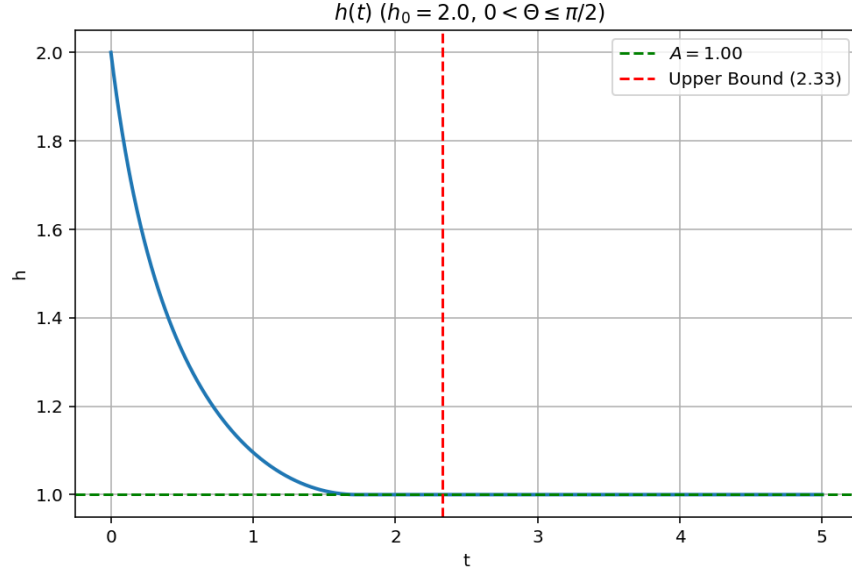
$$\begin{aligned}\kappa_h &= \frac{\sin(\theta)(2 - h^2 \sin(\theta)) + (1 - h^2)}{h(h + \sin(\theta)^2)^{3/2}} \\ &= \frac{\sqrt{1 - (A/h)^6}(2 - h^2 \sqrt{1 - (A/h)^6})}{h(h + \sqrt{1 - (A/h)^6})^{3/2}}\end{aligned}$$

Numerical Analysis

For $\pi/2 \leq \theta < 0$, we have

$$\frac{dh}{dt} = -\frac{1}{3}(h^6 - A^6)^{1/2}$$

Solving this with RK4, we get the plot



Points that reach the surface

At point $\theta = \pi/2$, we have

$$\frac{dh}{dt} = -h^3 \implies \frac{1}{h^2} - \frac{1}{h_0^2} = 2t$$

Thus as $t \rightarrow \infty$, we must have that $h \rightarrow 0$. Rearranging further

$$h^{-2} = 2t + \frac{1}{h_0^2} = \frac{2th_0^2 + 1}{h_0^2}$$

So

$$t \gg 1, \quad h \sim \frac{1}{\sqrt{2t}}$$

1 Problem with Ω starting at 0 and growing

Problem with $\Omega = \gamma t$

We have the characteristics

$$\frac{dh}{dt} = -\frac{1}{3}h^3 \sin(\theta) \quad \frac{d\theta}{dt} = \gamma t - h^2 \cos(\theta)$$

2 Poincare Compactification

Poincare Compactification pushes an $\mathbb{R}^n$ dynamical system into S^{n+1} . We get

$$(h, \theta) \rightarrow \left(\frac{h}{\sqrt{1+h^2+\theta^2}}, \frac{\theta}{\sqrt{1+h^2+\theta^2}}, \frac{1}{\sqrt{1+h^2+\theta^2}} \right) = (u, v, w)$$

We then get

$$\begin{aligned}\dot{u} &= \frac{u^3 v}{w} \cos\left(\frac{v}{w}\right) - \frac{1}{3} \left(u^3 + \frac{u^3 v^2}{w^2}\right) \sin\left(\frac{v}{w}\right) - uvw\Omega \\ \dot{v} &= \frac{u^4 v}{w^2} \sin\left(\frac{v}{w}\right) - \left(\frac{u^4}{w} + u^2 w\right) \cos\left(\frac{v}{w}\right) + (w^3 + u^2 w)\Omega \\ \dot{w} &= \frac{1}{3} \frac{u^4}{w} \sin\left(\frac{v}{w}\right) + u^2 v \cos\left(\frac{v}{w}\right) - w^2 v \Omega\end{aligned}$$